

Project Development Phase

Dataset Description & Data Dictionary

Field	Details
Team Lead	Govindu Kavya Sri
Team Members	Eswari Gayatri Budumuru, Kiladi Kalavathi, Adaka Mounika, Panditi Jyostna
Domain	Retail / Cosmetics Analytics
Tools Used	Python, Microsoft Excel, Tableau Public
Dataset	Cosmetics product dataset — 1,472 records, 11 attributes
Published Dashboard	public.tableau.com/app/profile/govindu.kavya.sri/viz/cosmeticproject_17872016666220/Dashboard1

Project Context

The cosmetics industry is one of the fastest-growing and highly competitive sectors, driven by changing consumer preferences, diverse skin types, brand innovation, and evolving pricing strategies. With increasing demand for personalized beauty products, companies rely on data-driven insights to understand market trends, consumer behavior, and product performance.

This project, “Cosmetic Insights: Navigating Cosmetics Trends and Consumer Behavior with Tableau,” focuses on analyzing a cosmetics dataset to uncover meaningful patterns and actionable insights. The dataset includes key attributes such as brand, product category, price, rank, and skin type suitability (Dry, Oily, Combination, Normal, and Sensitive).

This project is designed to help businesses and analysts understand cosmetics market trends and consumer preferences. By analyzing brand performance, pricing patterns, product distribution, and skin-type coverage, the dashboard enables stakeholders to identify high-performing products, target specific customer segments, and optimize marketing and product strategies.

Scenario 1: Monitoring Consumer Preferences

In a real-time scenario, imagine receiving an alert indicating a concerning trend in consumer preferences, such as a significant decline in interest in certain cosmetic products or ingredients. Using the Cosmetic Insights data, we can promptly assess the extent and potential impact of this trend, identify contributing factors, and deploy immediate interventions to adapt product offerings and marketing strategies. Whether it's through targeted promotional campaigns, adjustments in product formulations, or personalized

recommendations, real-time analysis enables agile decision-making and proactive measures to meet evolving consumer needs.

Scenario 2: Addressing Product Concerns

In the event of identifying widespread product concerns, such as negative reviews or safety issues associated with specific cosmetic items, real-time access to Cosmetic Insights data enables swift response and management. Cosmetic companies and regulatory bodies can utilize the dataset to gather crucial information about the concerns, including their prevalence, potential impacts on consumer trust, and affected product demographics. By leveraging real-time analytics, they can implement quality control measures, recall products if necessary, and communicate transparently with consumers to address their concerns and maintain brand integrity.

Scenario 3: Predictive Analysis and Product Innovation

Leveraging predictive analytics capabilities, Cosmetic Insights empowers companies to anticipate and respond to emerging trends and consumer preferences in the beauty industry. By analyzing historical data and identifying predictive indicators, companies can proactively innovate new products, adjust existing formulations, and tailor marketing strategies to meet evolving consumer demands. Real-time monitoring of market trends, consumer feedback, and competitor activities enables timely interventions, product innovation, and strategic decision-making to stay ahead in a competitive market landscape.

Dataset Overview

Field	Value
Dataset Name	cosmetics_cleaned.xlsx (cleaned extract)
Source	Cosmetics product dataset (cosmetics.csv)
Record Count	1,472 rows
Coverage	6 product categories · 116 brands

Data Dictionary

Field	Type	Description
Label	Text (Dimension)	Product category — Moisturizer, Cleanser, Face Mask, Treatment, Eye cream, Sun protect
Brand	Text (Dimension)	Manufacturer/brand name (title-cased during cleaning)
Name	Text (Dimension)	Product name
Price	Number (Measure)	Retail price (USD)
Rank	Number (Measure)	Average customer rating on a 1–5 scale; 0 converted to missing (unrated)

Field	Type	Description
Ingredients	Text (Dimension)	Full ingredient list as text; scrape-artifact placeholders converted to missing
Combination_Skin / Dry_Skin / Normal_Skin / Oily_Skin / Sensitive_Skin	Number (0/1)	Binary flags indicating skin-type suitability, later pivoted into Skin_Type / Suitable
Skin_Type (derived)	Text (Dimension)	Long-format skin type name, produced by pivoting the five binary flag columns
Suitable (derived)	Number (0/1)	Long-format suitability flag corresponding to each Skin_Type row

Cleaning & Preparation Summary

Brand names were standardized from all-uppercase to consistent title case. 143 rows of scrape-artifact ingredient text (placeholders such as “No Info” and “Visit the [Brand] boutique”) were converted to proper missing values. 19 rows with a Rank of 0 were treated as unrated rather than a real score, since Rank is otherwise a 1–5 scale. The five skin-type columns were renamed for clarity and later pivoted in Tableau into a long-format Skin_Type / Suitable field pair, expanding the working table from 1,472 to 7,360 rows for flexible filtering. Full step-by-step detail is documented in the Preprocessing Steps & Business Questions file in this same folder.

Known Limitations

- Ingredients text is unstructured; no ingredient-level parsing or active-ingredient extraction was performed in this phase of the project.
- Rank represents an aggregate customer rating rather than individual review counts, so products with very few reviews are weighted the same as those with many.
- The dataset reflects a snapshot of product listings rather than a live, continuously updated feed; flagged as a Future Scope item.